

Three Web Real-Time Multiplayer Games (Mobile/PC Browser) — MGG Arena Platform API Integration

This document is a Request for Proposal (RFP) for outsourcing the development of three web (WebGL) real-time multiplayer games.

We would appreciate it if you could submit a quotation that includes the cost and development timeline for each game.

0. Purpose & General Principles

0.1 Purpose

The goal is to develop three real-time PvP multiplayer games that run in browsers (mobile + PC) and deliver them in an operable state integrated with the **MGG Arena Platform API** (login, matchmaking, item purchases, rewards).

This scope also includes a **minimal Admin tool** to enable basic operations and management.

0.2 General Rule (Important)

For any requirements **not explicitly defined** in this RFP, the game should generally follow the **UX / rules / flow of the reference games**.

However, if there is any conflict with **legal/IP restrictions (no copying)** or this RFP, **this RFP takes precedence**.

1. Legal / IP / License Compliance (Mandatory)

1.1 Allowed Use of References

Reference games are used **only for benchmarking** genre feel, controls, and the match loop (game flow).

1.2 Prohibited (No Copying / No Look-alikes)

Copying any reference game's code or assets is prohibited, including:

- Recognisable map layouts
- Distinctive UI layouts / icons / typography / colour signatures
- Character/weapon silhouettes, names/logos, or audio signatures

1.3 Allowed (Asset Usage Policy)

Using **free assets** and **Unity Asset Store assets (including paid assets)** is generally allowed. However, all of the following conditions must be met:

- The license must permit commercial use, redistribution constraints, and platform deployment for this project.
- The developer must provide license proof when requested (purchase receipts, license links, permitted scope).
- Outputs that could be interpreted as “recognisable imitation of the reference game” (silhouette/layout/signature) are prohibited.

1.4 Must Be Included in the Quotation

- A non-infringement assumption statement
- A list of third-party libraries/assets including license type, links, cost, and usage scope

2. Project Overview (3 Games)

- **Project A:** Top-down arena brawler (reference-based) (*Brawl Stars style*)
- **Project B:** Arena shooter deathmatch (reference-based) (*Vortex9 style*)
- **Project C:** Mini battle royale (reference-based, minimized scope) (*Fortzone style*)

3. Common Requirements (All Projects)

3.1 Platform / Client

- Runs on **mobile + PC browsers (Web)**
- Target Devices: Galaxy S20 / iPhone 12 or equivalent (Mid-range reference).
- Target browsers: **Chrome / Safari (iOS) / Edge** (Firefox if feasible)
- Inputs:
 - PC: keyboard + mouse
 - Mobile: touch (virtual sticks/buttons)
- Responsive UI for both mobile and desktop layouts
- Performance target (guideline): playable smoothly on mid-range mobile devices and entry-level laptops (may propose target devices)

3.2 Real-Time Multiplayer + Bots

- Real-time multiplayer is required
- If there are not enough users, fill the match automatically with bots

- Party/friend invites: excluded
- Matchmaking: one simple random queue

3.2.1 Drop Handling (Required)

- Leavers are replaced by bots; the match continues

3.2.2 Reconnect (May Be Excluded if the Additional Effort Is Significant)

Requirements:

- Reconnect within a defined time window after network loss, app switching, or browser refresh
- Restore match state on reconnect (score/HP/inventory, etc.)
- If reconnect fails, the bot replacement remains and the match continues (the user is guided to the results screen)

3.3 Account / Login Policy

- No social login
- Accounts are handled via **platform account/token** (Platform API integration is mandatory below)
- Nickname policy follows platform rules; a basic client-side profanity filter must be applied

3.4 Progression / Meta (Scope Locked)

- No leveling/XP, missions/quests, mode unlocks
- No ads
- **Item purchases and rewards are mandatory** due to the Platform API requirements (see 3.10)

3.5 Character Skins & Lightweight Customisation

Included scope:

- Skin colour/tint changes (e.g., base colour + 3–5 variations)
- Lightweight part-based customisation (e.g., a small set of hats/helmets/glasses/backpacks)
- Selection UI (in lobby or character selection screen)

Identification rule (Required):

- In **Project B/C**, ensure each player/bot has a distinct visual identifier (e.g., tint/hat colour) to avoid identical-looking characters.

3.6 Maps (Cost-Minimisation Rule)

“Two maps” may be satisfied by:

- **1 layout + 2 theme skins** (e.g., default / snowy overlay)

3.7 Visual Direction Proposals (Required)

- Provide **at least 3 distinct visual direction proposals** for the full set of three projects
- Even when using free/Asset Store assets, consistent art direction and optimisation are required

3.8 Logging / Analytics (Minimum)

- Match start/end, join/leave, win/loss/rank, kills/deaths (or score)
- Basic client/server error logging

3.9 Security (Minimum)

- Where feasible, validate core combat values server-side (damage/fire rate/move speed), given real-time PvP
- Basic anti-tamper principles (e.g., do not trust client-side values; server-authoritative handling where applicable)

3.10 MGG Arena Platform API Integration (Mandatory)

Integration with the **Arena Game Server API** is mandatory, and must include all of the following:

- Login
- Matchmaking
- Item purchases
- Rewards (result submission and reward reflection)

3.10.1 Authentication / Session (Required)

- All requests must use **Bearer Token** authentication
- Token expiry/401: refresh and retry the original request (standard policy required)

3.10.2 Matchmaking / Game Start / Results / Rewards (Required)

- Before match start: call **Start Matching**
- On match start: call **Game Start** (including user/bot counts)
- On match end: submit **Game Result** and reflect the returned reward values in the UI
- Result submission must handle duplicate submissions / “already processed” responses robustly (consider idempotency)

3.10.3 Item Purchases (Required)

- Clearly display success/failure (insufficient balance/item not found, etc.) in UX
- Synchronise and display updated balance

3.10.4 Operations Data Sync (Required)

- Rooms / Reward settings / Items must be retrievable and applicable via the Platform API
- The developer must provide an **initial seed template + operations guide**

3.10.5 Platform API Logging / Security (Required)

- Logs must be traceable via correlation keys (e.g., room_id/sequence)
- Mask tokens/sensitive data in logs

3.11 Audio & SFX Requirements (Immersive Standards)

- Sound Count: Minimum 30+ unique SFX per game (UI, Ambient, Combat)

3.12 Admin Tool–Based Operations/Management (Minimum)

This project includes a minimal scope for an **Admin tool** to enable basic operations and management.

The following functions are the **minimum required**. Advanced operations features (analytics dashboards/CS tools/granular permissions) are out of scope.

3.12.1 Access / Security (Required)

- Provide an admin page at a separate URL (e.g., /admin)

- Admin authentication must be implemented (username/password or a platform-provided method)
- Minimum permission model: a single Admin role with full access is acceptable

3.12.2 Game Management via Admin Page (Minimum Required)

(1) Game session / match management

- View a list of active/queued games (sessions/rooms)
- View session details (room ID/sequence, player count user/bot, start time, status)
- Provide at least one minimal action (choose one):
 - a) Force end a session (Abort/Close), or
 - b) Disable a session state (Disable), or
 - c) Pause/Resume matchmaking for a room (Pause/Resume)

(2) Operations configuration management (Platform API-synced)

- Rooms / Reward settings / Items must be viewable in the admin page
- Provide minimal controls such as an Enable/Disable toggle and/or a Refresh (re-sync) button
- The Platform API (3.10.4) remains the source of truth for persistence/permissions/validation.
The admin page may be implemented as a “thin UI” that calls the Platform API (minimum implementation).

(3) Incident response (Minimum)

- Provide at least one screen to check recent error logs (last N) or API failure counts (a simple table is sufficient)

3.12.3 Explicitly Out of Scope (Admin Tool)

- CS ticket / report workflow
- Advanced analytics / retention / revenue dashboards
- Granular RBAC, SSO, and audit logs
These are excluded from this scope.

3.13 Game Feel & Polish Requirements (Mandatory)

To ensure high quality without excessive asset costs, the following "Juice" elements must be implemented

- **Visual Feedback:**
 - **Screen Shake:** Subtle camera shake on heavy attacks, explosions, and taking damage.

- **Damage Floaters:** Floating text showing damage values (color-coded: White for normal, Red/Yellow for critical).
- **Hit Flash:** Enemies must flash white/red briefly (0.1s) when hit.
- **Audio Feedback:**
 - **Impact Sounds:** Distinct sounds for hitting an enemy vs. hitting a wall.
 - **Low Health Warning:** Audio/visual vignette effect when HP is below 30%.

4. UI Screen Inventory (Required)

4.1 Required Screens (Common)

- Boot / Loading (preload)
- Title / Start
- Nickname setup (+ profanity filter)
- Main Lobby
 - Play
 - Skins/Customisation entry (included by default)
 - Settings
 - Balance display
- Settings (overlay or separate)
 - Audio (BGM/SFX)
 - Controls (sensitivity / mobile layout)
 - Graphics (quality toggle)
- Matchmaking Queue (with cancel)
- In-game HUD (HP/ammo/skills/score/crosshair as relevant)
- In-game Scoreboard
- Pause Menu (resume / exit)
- Match Results (win/loss/rank + 3–5 stats; play again/return)
- Error / Disconnect (retry / return)
- Reconnect UI

4.2 Additional Screens

- Tutorial (one-page or in-game hints)
- Report/Block (moderation)

4.3 Admin Screens

- Admin Login
- Game Session List / Detail
- Rooms/Reward/Items Sync View (view + refresh button)
- Error/Status View (recent errors / failure counts)

5. Per-Game Requirements

A) Project A — Top-Down Arena Brawler

A-1 Game Modes: 2

- 3v3 Team Mode
- 6-player FFA Mode

A-2 Characters: 5

- Fixed at 5 characters
Use standard MOBA archetypes to simplify balancing and logic implementation:
- 1. **The Fighter (Balanced):** Mid-range shotgun/rifle, moderate HP. (Ref: Shelly)
- 2. **The Tank (Melee):** High HP, short-range punch/smash. (Ref: El Primo)
- 3. **The Sniper (Long Range):** Low HP, very long range, high damage single shot. (Ref: Colt/Piper)
- 4. **The Thrower (AOE):** Lobs projectiles over walls. (Ref: Dynamike)
- 5. **The Assassin (Speed):** High move speed, quick burst damage. (Ref: Crow/Leon)
- Each character: Basic attack + Active skill + Ultimate (3 skills)

A-3 Match Length / Rules

- Follow the reference game flow (RFP takes precedence if there is any conflict)
- Match length guideline: 3–5 minutes

A-4 Maps

- Base: 1 layout + 2 theme skins (default/snow)
- Optional: an additional unique layout (priced separately)

A-5 Art/Audio (Minimum)

- Animations: idle/move/attack/hit/death
- Minimal skill VFX
- BGM: 1 lobby, 1 in-game
- SFX: to be defined

B) Project B — Arena Shooter Deathmatch

B-1 Game Modes: 2

- 4v4 Team Mode
- 8-player FFA Mode

B-2 Players/Bots

- Team mode: total 8 (4v4), bots fill missing slots
- FFA mode: total 8, bots fill missing slots

B-3 Weapons

- 7+ weapons (including grenades)

B-3.1 Weapon Classes The 7+ weapons must define the meta, not just duplicate stats.

1. **Melee:** Bat or Knife (Fast swing).
2. **Pistols:** Standard Handgun, Heavy Revolver (Slow but high dmg).
3. **Automatics:** SMG (High fire rate, low accuracy), Assault Rifle (Balanced).
4. **Heavy:** LMG (Large mag, slow move speed).
5. **Precision:** Sniper Rifle (Scope zoom required, 1-2 shot kill).
6. **Explosives:** RPG/Bazooka (Splash damage), Grenade Launcher.
7. **Special:** Shotgun (Close range burst).

B-3.2 Weapon Mechanics (Addendum)

- **Recoil & Spread:** All automatic weapons must have procedural recoil (camera moves up) and bullet spread logic to ensure shooting feels realistic, not static.

B-4 Maps

- Base: 1 layout + 2 themes (default/snow)
- Optional: an additional unique layout (priced separately)

C) Project C — Mini Battle Royale (Minimized Scope)

C-1 Max Players: 20

- Recommended: up to 20, bots fill missing slots

C-2 Core Loop (Required)

- Drop from plane
- Looting (weapons/ammo/heals)
Auto-Loot: Ammo and health packs are automatically picked up when walked over (if not full).
Tap-to-Equip: Weapons on the ground are picked up via a simple interact button or tap.
- Shrinking safe zone (2–4 phases)
- Final survivor/team

C-3 Weapons

- 10+ weapons

C-4 Weapon Attachments/Parts Included

- Example: scope/magazine/grip/suppressor (minimum 1–2 types)
- Avoid excessive complexity

C-5 Maps

- Base: 1 layout + 2 themes (default/snow)
- Optional: an additional unique layout (priced separately)

C-6 Vehicles

- 2 vehicle types A. Fast Car (Sports/Buggy), B. Heavy Truck (Slower, High Cover)
Physics: Basic arcade physics (no realistic simulation needed, but must not flip over too easily).

C-6 Exclusions (High-Cost Features)

- Building/fortification excluded
- Destructible environment excluded
- Duo/squad or multiple modes excluded

6. Deliverables

6.1 Infrastructure & Deployment (Client-Hosted)

- **Target Environment:** All server-side components must be deployed to the Client's provided cloud account (AWS/GCP - TBD).

Deliverables:

- Full Source Code (Client/Server).
- **IaC (Infrastructure as Code) Scripts:** Dockerfiles, Kubernetes manifests, or Terraform scripts required to replicate the environment.
- **One-Click Deployment Guide:** Documentation allowing the Client to redeploy the service from scratch without Vendor assistance.

Ownership: The Client retains full ownership of all data, logs, and infrastructure configurations upon handover.

6.2 Documentation

- Technical documentation (architecture, configuration, operations points, troubleshooting)
- Platform API integration documentation (endpoints, edge cases, retries, idempotency)

6.3 Assets / Licenses

- Asset list + license inventory (including Asset Store)
- Original asset sources where feasible (to be agreed)

6.4 QA

- Basic QA report (test cases, bug list, fixes)

6.5 Operations Sync (Required)

- Rooms / Reward settings / Items sync integration guide
- Initial seed template + operator manual

7. Acceptance Criteria

7.1 Common E2E

- Full flow works on mobile + PC browsers:
Load → Lobby → Matchmaking → In-game → Results
- Bots fill reliably; matches start and end properly
- Two themes (default/snow) are verifiably applied
- No critical issues at release (crash / infinite load / cannot finish match)

7.2 Reconnect

- Reconnect succeeds within the defined time window after network loss
- Even if reconnect fails, the match remains stable (bot replacement), and the user receives a clear results/return UX

7.3 Platform API E2E (Required)

- The following flow passes for all three games:
Login → token verify/refresh → balance inquiry → start matchmaking → game start → submit result → reward reflection → item purchase
- Stable handling under edge cases (insufficient balance, duplicate submissions, token expiry, etc.)

7.4 Admin Tool Acceptance (Minimum)

- Admin login works and access control is enforced
- Session/room list and detail view work
- The chosen minimal action (Abort/Close OR Disable OR Pause/Resume) works as intended
- Rooms/Reward/Items view and refresh/re-sync works (via Platform API integration)
- Error/status screen shows recent failures (or recent error logs)

8. Warranty & Support

- **Warranty Period:** 6 months from the date of final acceptance (Go-live)
- **Scope:** Fixes for critical bugs (crashes, logic errors, sync failures) and security vulnerabilities found post-launch.
- **Response Time:** Critical issues (server down, payment fail) must be addressed within 24-48 hours.
- **Exclusion:** Feature additions or design changes are excluded from warranty but may be contracted separately.

9. Quotation Submission Requirements

- Cost per game (A/B/C)
- Timeline and milestones: planning/design/development/integration/QA/release
- Multiplayer architecture proposal: server authority, sync method, bot design, load handling
- Platform API integration plan: endpoint scope, edge cases, retries/idempotency, test plan
- Asset/license plan: free/Asset Store usage list and compliance approach
- Clearly list out-of-scope items and assumptions
- Risks and mitigations (performance, cheating, latency, browser compatibility, operations data sync)